Pediatric Vaccination



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Vaccine is a substance that stimulates the body's immune response.

Outline of Immunization procedures

Active Immunity	Mode of action	Example
Live attenuated vaccines	• Attenuated (weakened) form of the "wild" virus or bacterium • Immunogenic but not pathogenic (cannot induce the full blown disease) • Live organism multiply in the host and the resulting antigenic dose is larger than what is injected • Live vaccines have all the major and minor antigenic components • Severe reactions possible. • Fragile – must be stored and handled carefully. • Usually produce immunity with one dose (Except the oral vaccines)	1. MMR 2. BCG 3. SABIN(Oral polio) 4. VARICELLA ZOSTER 5. Rota virus 6. YELLOW FEVER 7. Influenza(intranasal) 8. Typhoid
Killed, inactivated vaccines	• killed by heat or chemicals. • Cannot replicate in the host. • Generally not as effective as live vaccines. • Antibody titers may diminish with time. • Usually safe. • Usually require multiple doses to produce adequate antibody response.	1. pertussis, 2. IPV 3. Influenza: intramuscular 4. Rabies 5. Typhoid 6. Hep A 7. cholera 8. plague

Toxoid	• Certain organisms produce exotoxins. • Toxins are detoxicated and used in the preparation of vaccine. • Toxoid vaccines are highly effective and safe immunizing agents.	Diphtheria, Tetanus
Subunit Vaccines	• Antibody with less functional activity. • Immunogenicity improved by conjugation.	1. Capsular polysaccharides (Pneumococcal vaccine). 2. Cell wall polysaccharides (Meningococcal vaccine)

Other types:

Recombinant vaccine • It is prepared by cloning genes for determined antigens in yeast cells where it is expressed. • Then used for vaccine preparations. • Genes for determined antigens are inserted into a vector, usually a virus, that is not dangerous. • as HepB vaccine, HPV vaccine.	**Combinations vaccines** • More than one type of immunising agent, as: MMR, DPT, DT, Hexa. • Combined to: 1. simplify administration. 2. reduce costs . 3. minimize the number of contact of the individual with the health system.
Polyvalent vaccine: more than one strain of the same species. • Oral polio vaccine (1,3), • Meningococcal vaccine (A,C,Y,W135), • Rota vaccine • Pneumococcal vaccine (13 serotypes).	**Conjugated vaccine:** • A special type of subunit vaccine, • Made by linking to a carrier protein to enhance the immune response. • as Meningococcal vaccine.

Vaccines used in Libya by classification age of administration. Route and dose for routine NPI

Age of administration	Vaccine	Dose	Route
At birth	BCG OPV Hep B	0.05 ml 2 drops 0.5 ml	ID Oral IM
2 months	Hexa vaccine =(DPT + Hep B+Hib + IPV) Rota vaccine PCV	0.5 ml 2ml 0.5 ml	IM Oral IM
4 months	Hexa vaccine =(DPT + Hep B+Hib + IPV) Rota vaccine PCV	0.5 ml 2ml 0.5 ml	IM Oral IM
6 months	Hexa vaccine =(DPT + Hep B+Hib + IPV) Rota vaccine	0.5 ml 2ml	IM Oral
9 months	Meningococcal vaccine (ACYW135) OPV	0.5 ml 2 drops	IM Oral
12 months	Meningococcal vaccine (ACYW135) MMR vaccine PCV = pneumococcal vaccine	0.5 ml 0.5 ml 0.5 ml	IM SC IM
15 months	Chicken pox vaccine Hep A	0.5 ml 0.5 ml	IM IM
18 months	Penta vaccine =(DPT+Hib + IPV) MMR vaccine	0.5 ml 0.5 ml	IM IM
6 years	Tetra vaccine = (DPT vaccine +IPV) Chicken pox vaccine	0.5 ml 0.5 ml	IM IM
12 years	Meningococcal vaccine (ACYW135) HPV (human papillomavirus vaccine)	0.5 ml 0.5 ml	IM IM
15 years	TdaP	0.5ml	IM

Contraindications to any routine active immunization:

- Acute febrile illness, malaise, cough, diarrhea or other symptoms requiring medical treatment.
- Severe local or general reaction to preceding dose of the same vaccine.

Contraindication to the use live attenuated vaccine:

- Immune deficiency: Humoral, cellular or bot / 1ry or 2ry as HIV.
- Malignant conditions such as Leukemia or Hodgkin's disease.
- Immunosuppressant: as BM transplantation within 6 months, Chemotherapy or radiotherapy within 6 months ,large prolonged dose of CS.
- Pregnancy
- Severe protein energy malnutrition "only if very severe PEM"

Conditions which are **Not** contraindication of vaccination:–

1. Common cold in a mild respiratory system.
2. Mild local reaction to a previous dose.
3. Family history of allergy, eczema or asthma
4. Personal history of eczema except for smallpox vaccine which is not now used.
5. An antibiotic course.
6. Prematurity or preterm delivery Or IUGR.
7. Breast feeding
8. Congenital heart disease chronic disease of chest e.g: cystic fibrosis
9. Down syndrome
10. F/H of SIDS.

BCG

- One dose soon after birth , Intradermal route.
- Dose:0.5ml (0.05ml if less than 1 month)
- Site: about one third down the left upper arm over the insertion of the deltoid muscle.
- The vaccine is 70–80% effective against the most severe forms of TB (TB meningitis in children). It is less effective in preventing respiratory disease (more common in adults).

Contra indication of BCG vaccine:

- Tuberculin positive as assessed by the mantoux test .
- Patients with HIV infection whether symptomatic or asymptomatic
- Other contraindications of live attenuated vaccine as mentioned .

Complication of BCG vaccine:

- Persistent ulcer at site of vaccination (scare)
- Regional suppurative lymphadenitis

- Marked lymphadenitis with cold abscess and tuberculous sinus
- BCG-oma if given SC. injection
- Disseminated TB infection especially in infants with immunodeficiency.

Infants with no scar should be revaccinated ?

- A scar will appear in most people vaccinated with BCG but not everyone.
- If the child didn't have this reaction it doesn't mean that they have not responded to it.
- Tuberculin skin test should be carried out. If negative : give a second BCG vaccination. This revaccination is given in the RIGHT arm. It's not necessary to inspect this scar. Advise not to have any vaccination in the right arm for three months after BCG

polio vaccine

Live attenuated (OPV , Sabin)

- oral route. Uphold herd immunity by cross infection
- At birth, 9months .

OPV may cause vaccine associated paralytic polio

Inactive Polio Vaccine (IPV, Salk)

- Inactivated polio vaccine, given IM.
- as a part of the Hexa or penta vaccine

OPV or IPV?

- WHO no longer recommends an OPV-only vaccination schedule.
- For all countries currently using OPV only, at least 1 dose of IPV should be added to the schedule
- WHY? Vaccine associated polio VS high risk in of transmission from neighboring countries

Contraindication for oral polio vaccine: as CI of any live attenuated vaccine

- Immune – compromised, including those congenital or acquired immunodeficiency.
- OPV also should not be given to household contact immunocompromised individuals or to subsequent siblings of child with congenital immunodeficiency unital the younger child is shown to be normal.
- OPV should be delayed in individuals receiving immunosuppressive therapy until at least 3 months after stopping the therapy.

DPT: (Diphtheria, pertussis, Tetanus)

- killed the vaccine and toxoid .
- 0.5 ml IM or S.C in 3 doses & 3 boosters.

Contraindications are:–

1. An acute febrile illness.
2. An evolving or suspected neurologic disease.
3. Severe reaction to prior dose of DPT.

S.E:

- local swelling and tenderness at injection site, slight fever,irritability, unusual shock like syndrome(uncommon)and convulsions or manifestations of encephalopathy (rare, 1:500000)
- Now replaced by HEXA: DaPT +Hib + HepB + IPV.

Pertussis

- Whole cell of killed organism DTP :– Pertussis in vaccinated person is mild and short duration
- Acellular pertussis vaccine Tdap :– For who can not tolerate whole cell, also it has less side effects

S/E (common)	Contraindication	Not Contraindication
Pain , redness ,swelling at injection site Fever >38 c Drowsiness Anorexia and or vomiting	1. History of post vaccination anaphylaxis 2. Anaphylaxis to neomycin, streptomycin, polymyxin 3. Uncontrolled convulsion 4. Progressive neurological illness 5. Acute febrile illness, not minor	1. F/H of epilepsy 2. Antibiotic treatment 3. Past History of febrile convulsion 4. Stable cp

HBV Vaccine

What is the current recommendation for a baby born to a mother with hepatitis B ?

- Infants born to mothers who are hepatitis B surface antigen–positive are recommended to receive both hepatitis B vaccine and HBIG.Both should be given on the day of birth, at the same time but in separate thighs .
- Infants should receive 3 subsequent doses of hepatitis B vaccine at 2 , 4 , 6
- months of age , so that they receive a total of 4 doses of hepatitis B vaccine.
- The dose of HBIG (hepatitis B immunoglobulin) is 100 IU given by intramuscular injection

What is hepatitis B immunization management of preterm infants weighing < 2000 g ,by maternal hepatitis B surface antigen negative ?

- If negative / Delay first dose of hepatitis B until age 1 month or hospital discharge and complete the hepatitis B vaccine series at ages 2 , 4 , 6 months.

MMR

- Recommended 2 doses/ at 12,18 months .
- SC route.
- Possible side effects:
    a. Soreness and erythema
    b. Fever and rash (or mild form of the disease 7 to 10 days later)

Contraindication of MMR vaccine:

- Pregnancy– because of the risk of fetal damage.
- Anaphylactic reaction to egg ingestion (vaccinated after skin test)
- Anaphylactic allergy to neomycin (Because vaccine contain neomycin)
- Immunoglobulin within 3 months because of blunt immune response.
- Immunocompromised except HIV on ART – because of severe symptomatic infection

Rotavirus vaccine:

- Live attenuated virus. Recommended at 2, 4, 6 months.
- Rotarix : 2–dose series at age 2 and 4 months
- Rotateq : 3–dose series at age 2, 4, 6 months
- Oral route.(2 drops)
- Side effects may include diarrhoea, fever....

Precautions of Rotavirus vaccine administration :
Co–administration of the oral polio and rotavirus vaccines may be accompanied by an interference on immunogenicity.

Pneumococcal vaccine:

- 13–valent pneumococcal conjugate vaccine.
- Given at 2, 4, 12 months.
- It is used to prevent pneumonia,meningitis, and sepsis caused by various types of pneumococcal bacteria Streptococcus pneumoniae polysaccharides serotypes 1, 3,4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A,19F and 23F bacterial
- 0.5 mL dose IM

S.E: >10%
Fever, Irritability,Decreased appetite, Redness, swelling, and tenderness at injection site

Indications:

- Individuals 2 years of age and older with HIV, Lymphoma , Hodgkins disease , Generalized malignancy , Chronic renal failure , Nephrotic syndrome , Receiving immunosuppressive therapy , organ or bone marrow transplant , Anatomic or functional asplenia, Sickle cell disease , Cystic fibrosis , Diabetes , Cochlear implant , Chronic neurological conditions that may impair clearance of oral secretion
- Adults 65 years of age and older

Hepatitis A vaccine

- Indicated as routine vaccination for children at least 12 months of age (at 15 months of age)
- <12 months: Not indicated/ if administered, not counted toward the routine 2-dose series
- Primary dose: 0.5 mL IM
- Administer booster dose 6–18 mo after primary dose

Who should get the vaccine ?

- Children over 1 yrs ·
- Hep A lab workers ·
- People travelling to areas with high incidence of Hep A .
- Chronic liver disease patients.
- Drug-abusers .

Human papilloma (HPV) vaccine

- Provides protection against the two strains (human papillomavirus 16 & 18) that cause 70% of cervical cancer. 2–3 doses of vaccine are given over a 6-month period to girls aged 12–13 years of age
- Recombinant DNA, IM .

VARICELLA VACCINE

- Dose, route: – 0.5ml SC OR IM.
- Site: Anterolateral thigh or upper arm
- Schedule: 15 months,6 years
- Adverse reactions: Fever, rash, local pain or redness
- Contraindications: Anaphylaxis after previous dose, lymphopenia, immunodeficiency ,during immunosuppressive therapy
- Storage: 2–8°C/ protect from light/ use within 30 min of reconstitution.

Influenza vaccine

- 0.5 ml – IM .

People who are :

- are a child aged 2 to 12 years
- are pregnant
- all those aged 6–23 months and 13–64 years at increased risk of influenza

Related complications such as:

(Those with chronic illness ,e.g. chronic heart disease (including acute coronary syndrome) chronic liver disease , chronic neurological disease (where the neurological condition compromises clearance of respiratory secretions), chronic renal failure , chronic respiratory disease (including chronic obstructive pulmonary disease , cystic fibrosis , moderate or severe asthma , and bronchopulmonary dysplasia), diabetes mellitus , or haemoglobinopathies).

- Are 65 years of age and over.

The cold chain

- Cold Chain is the strict system (people and equipments) used for storage and distributing vaccines in good condition.
- Vaccines are sensitive to heat and freezing and must be kept at the correct temperature from the time they are manufactured until they are used.

Which do you never want to freeze?

- OPV, Measles, MMR, BCG can be frozen, but not diluents.

Sensitivity to light

- BCG, measles and MMR vaccines are sensitive to light.
- Normally, these vaccines are supplied in vials made from dark brown glass.
- Should be kept away from direct light.

Expanded program of immunization EPI

- Established 1974 by the WHO.
- Its aim was to eradicate/reduce the incidence of common and often Fatal diseases.
- The six traditional targets were diphtheria, pertussis, tetanus, poliomyelitis, measles and tuberculosis.
- EPI can be expanded so other diseases may be targeted according to individual needs.